Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 21: Approved Solutions

Cardinality and Countability

Covers Chartrand & Zhang, Section 5.5

Status of this set

Ungraded practice. Solutions released with it. Shorter than usual, since Section 5.5 had one period rather than three.

Part I: How to Practice This Material

Everything reduces to one question

Is there a bijection? $|A| = |B|$ exactly when one exists. Denumerable means a bijection with $\mathbb{N}$ — equivalently, the elements can be *listed* with each appearing exactly once.
To show a set is denumerable: **produce a listing**. To show it is uncountable: argue that *no* listing can work, which usually means a diagonal argument.

Three mistakes to expect

1. **Assuming a proper subset must be smaller.** True for finite sets, false for infinite ones.
2. **Confusing "infinite" with "uncountable."** $\mathbb{Z}$ and $\mathbb{Q}$ are infinite and countable.
3. **Trying to disprove a listing by testing it.** You must rule out *all* listings.

Part II: Practice Problems

Section A — Countable or not?

1. Classify each as finite, denumerable, or uncountable, with a one-line reason.

(a) The set of even integers.

(b) $\{x \in \mathbb{R} : 0 < x < 1\}$.

(c) The set of integers divisible by 100.

(d) $\mathcal{P}(\mathbb{N})$.

Approved solution

(a) Denumerable. List them $0, 2, -2, 4, -4, \ldots$ — the same zig-zag as for $\mathbb{Z}$. Equivalently, $f(n) = 2n$ is a bijection from $\mathbb{Z}$, which is denumerable.
(b) Uncountable. It contains an uncountable subset — essentially all of $[0, 1]$ — so Theorem 5.81 applies. (A diagonal argument works directly too.)
(c) Denumerable. List $0, 100, -100, 200, -200, \ldots$ Every infinite subset of a denumerable set is denumerable (Theorem 5.78), and this is an infinite subset of $\mathbb{Z}$.
(d) Uncountable. By Theorem 5.86, $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$, so the power set cannot be denumerable.

(a) and (c) are the instructive pair. Both are proper subsets of $\mathbb{Z}$ that look far smaller — one is "half" and the other "one hundredth" — and both have exactly the same cardinality as $\mathbb{Z}$ itself. Thinning an infinite set does not shrink it.

Section B — Listings

2. Give an explicit listing showing that the set of odd integers is denumerable, and state the bijection it defines.

Approved solution

List: $1, -1, 3, -3, 5, -5, \ldots$
Every odd integer appears exactly once, so the listing is a bijection $f : \mathbb{N} \to$ (odd integers) sending n to the nth entry.
Explicitly: $f(n) = n$ when n is odd, and $f(n) = 1 - n$ when n is even. Check: $f(1) = 1$, $f(2) = -1$, $f(3) = 3$, $f(4) = -3$. ✓
Any listing hitting every odd integer exactly once is acceptable — the formula is a bonus.

Section C — The diagonal argument

3. In Cantor's proof, the constructed number b has nth digit equal to 2 if $a_{nn} = 1$ and 1 otherwise.

(a) Given the list $a_1 = 0.5231\ldots$, $a_2 = 0.1104\ldots$, $a_3 = 0.9817\ldots$, find the first three digits of b.

(b) Why must b differ from every a_n?

(c) Why does the rule avoid the digits 0 and 9?

Approved solution

(a) The diagonal digits are $a_{11} = 5$, $a_{22} = 1$, $a_{33} = 1$. Applying the rule: $b_1 = 1$ (since $5 \neq 1$), $b_2 = 2$ (since $a_{22} = 1$), $b_3 = 2$. So $b = 0.122\ldots$
(b) By construction $b_n \neq a_{nn}$ for every n, so b and a_n differ in their nth decimal place. Two numbers differing in some decimal place are different numbers. Since this holds for every n, b is not on the list.
(c) Some numbers have two decimal expansions — $0.5000\ldots = 0.4999\ldots$ — so differing in a digit does not always mean differing as numbers. By using only 1s and 2s, b has no trailing 0s or 9s and therefore only one expansion, which closes that gap.

Part (c) is the subtle one and the reason the rule looks arbitrary. It is not a trick — it is repairing a genuine hole in the argument.

Section D — Writing

4. In five to eight complete sentences, explain why $\mathbb{Z}$ and $\mathbb{N}$ have the same cardinality even though $\mathbb{N}$ is a proper subset of $\mathbb{Z}$. Say why this cannot happen for finite sets, and connect your answer to a result from Week 7.

Approved solution

Model answer. Two sets have the same cardinality when a bijection between them exists, and for $\mathbb{N}$ and $\mathbb{Z}$ one does: the listing $0, 1, -1, 2, -2, \ldots$ pairs each positive integer with exactly one integer and reaches every integer. So $|\mathbb{N}| = |\mathbb{Z}|$, even though $\mathbb{N}$ sits strictly inside $\mathbb{Z}$. For finite sets this is impossible, because a one-to-one function from a finite set into a proper subset would have to squeeze n distinct images into fewer than n places. That is exactly the argument behind Practice Set 19 Problem 6, where a one-to-one function from a finite set to itself had to be onto — and the proof used finiteness in the step where a subset of equal size must be the whole set. Infinite sets are precisely the ones where that step fails, which is why $f(n) = 2n$ can map $\mathbb{Z}$ one-to-one into a proper subset of itself.

Full credit requires the connection to where finiteness was used. An answer that only exhibits the listing earns about half.

Section E — Challenge (optional)

5. Cantor's proof of $|A| < |\mathcal{P}(A)|$ builds $A_g = \{a \in A : a \notin g(a)\}$ and asks whether $a_0 \in A_g$ where $g(a_0) = A_g$. Explain why both answers lead to a contradiction, and say what this resembles from Lecture 1.

Approved solution

Suppose $a_0 \in A_g$. By the definition of A_g, membership requires $a_0 \notin g(a_0)$. But $g(a_0) = A_g$, so this says $a_0 \notin A_g$ — contradicting what we assumed.
Suppose instead $a_0 \notin A_g$. Since $g(a_0) = A_g$, that says $a_0 \notin g(a_0)$, which is exactly the condition for membership in A_g. So $a_0 \in A_g$ — again a contradiction.
Both answers fail, so no such a_0 exists, so A_g is not in the range of g and g is not onto. Since g was an arbitrary function $A \to \mathcal{P}(A)$, no bijection exists.

This is the Liar sentence from Lecture 1: "this sentence is false," which we classified as not a statement because assuming either truth value produced a contradiction. Here the same self-reference is a proof device rather than a curiosity, and it establishes that there is no largest cardinality.
The same shape returns as the halting problem in theory of computation.

Instructor notes

- **Problem 1(a) and 1(c)** together make the point that thinning an infinite set does not shrink it. Worth naming aloud.
- **Problem 3(c)** explains the 1/2 digit rule, which otherwise looks arbitrary. Strong students ask about it; this is the answer.
- **Problem 4** is the writing anchor and points back at where finiteness was used in Practice Set 19.

- **Problem 5** closes the Lecture 1 loop. If you have time for one thing from this set on the board, make it this.
- **Set is deliberately short** — Section 5.5 had one period, and the midterm is the following week.